MediFlow - Hospital Management System

MediFlow

AI-Driven Hospital Operations Management

Smarter Decisions for a Healthier Tomorrow

- Bed & Ward Allocation
- Patient Priority
- Doctor Recommendation
- Medicine Stock Alert



Login to MediFlow

Enter your credentials to access the system

admin

••••••••

Login

or

Authorized personnel only

This system contains confidential hospital data.

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- Dashboard
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- Settings
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Dashboard

Overview of hospital resources and recent activities.

Tuesday, 24 September 2025
12:10 PM

Welcome, Admin
Administrator

Total Patients
128
12% from last week

Available Beds
24 / 120
20% available

Total Doctors
18
4 on leave

Low Stock Medicines
7
Need restocking

Recent Patients

View All

ID	Name	Age	Condition	Priority	Status
P001	Rahim Uddin	45	Chest Pain	Urgent	Admitted
P002	Samira Akter	28	Fever	Normal	Observation
P003	Abdullah Khan	60	Breathing Issue	Emergency	ICU
P004	Nusrat Jahan	35	Pregnancy	Normal	Admitted
P005	Tanvir Hasan	22	Stomach Pain	Normal	Discharged

Quick Actions

- Add New Patient
- Bed Allocation
- Doctor Recommendation
- Check Medicine Stock
- View Resource Status
- Predict Resource Demand

Current Resource Status

View Details

Bed Occupancy
96 / 120
80% occupied

Doctor Availability
18 / 22
4 on leave

Medicine Stock
43 / 50
7 low stock items

AI Insights & Alerts

View All

Low Stock Alert — 24 Sep 2025
Paracetamol, Metformin and Saline are below minimum level.

Higher Patient Inflow Expected — 24 Sep 2025
Based on recent data, 20% increase in patient admission is predicted for next week.

Suggested Bed Allocation — 24 Sep 2025
3 patients can be shifted to General Ward based on current condition.

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Patient Priority Management

Enter patient information to get AI-assisted priority suggestion.

Patient Information

Patient ID: P006

Name: Mahfuz Ahmed

Age: 50 years

Gender: Male

Symptoms: Severe chest pain, shortness of breath

Existing Condition: Heart Disease

Suggest Priority (AI) | Clear

AI Suggestion Result

Suggested Priority:

Emergency

Reason:

Symptoms indicate possible cardiac issue and immediate medical attention is required.

Recommended Action:

Admit to ICU and assign a cardiology specialist.

Patient List

ID	Name	Age	Condition	Priority	Status	Action
P001	Rahim Uddin	45	Chest Pain	Urgent	Admitted	View
P002	Samira Akter	28	Fever	Normal	Observation	View
P003	Abdullah Khan	60	Breathing Issue	Emergency	ICU	View
P004	Nusrat Jahan	35	Pregnancy	Normal	Admitted	View
P005	Tanvir Hasan	22	Stomach Pain	Normal	Discharged	View

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Bed & Ward Allocation

Find suitable bed/ward for a patient using AI recommendation.

Patient Details

Patient ID: P001

Name: Rahim Uddin

Age: 45 years

Gender: Male

Condition: Cardiac

Priority: Urgent

Special Requirements: Needs monitoring

Suggest Bed/Ward (AI) | Clear

Available Beds

Ward: All | Type: All

Bed ID	Ward	Type	Status
B101	General	Normal	Available
B102	General	Normal	Occupied
B103	General	Normal	Available
B201	ICU	Critical	Available
B202	ICU	Critical	Occupied
B301	HDU	Normal	Available
B302	HDU	Normal	Available
B401	Private	Normal	Available
B402	Private	Normal	Occupied

AI Recommendation Result

Suggested Bed:

B201
ICU Ward

Ward Type: ICU (Critical)

Reason: Patient has cardiac condition and requires close monitoring.

Alternative Option: B301 (HDU)

Additional Note: Keep under observation and assign a cardiologist.

Assign This Bed

Ward Occupancy Overview

General Ward
15 / 60 beds — 25%

Available Beds:
B101, B103, B104, B105,...

Occupied Beds:
B102, B106, B107,...

ICU Ward
18 / 20 beds — 90%

Available Beds:
B201, B203

Occupied Beds:
B302, B204, B205, B206,...

HDU Ward
8 / 20 beds — 40%

Available Beds:
B301, B302, B303, B304,...

Occupied Beds:
B305, B306,...

Private Ward
6 / 20 beds — 30%

Available Beds:
B401, B403, B404,...

Occupied Beds:
B402, B405,...

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Doctor Recommendation

Get AI-assisted doctor recommendation based on patient needs.

Patient Information

Patient ID: P003

Name: Abdullah Khan

Age: 60 years

Gender: Male

Symptoms: Breathing issue, cough, chest pain

Preferred Department: General Medicine

Priority: Emergency

Recommend Doctor (AI) | Clear

AI Recommendation Result

Recommended Doctor:

Dr. Faisal Rahman

Cardiology Specialist

Reason: Symptoms indicate possible cardiac issue. Cardiology department is most suitable.

Availability: Available Today

Preferred Time: 10:00 AM - 12:00 PM

Department: Cardiology

Alternative Option: Dr. Nusrat Ahmed (Pulmonology)

Available Doctors

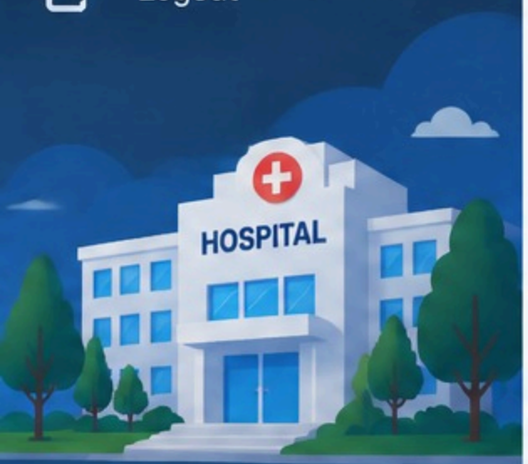
Search by name or specialization... | All Departments | Available Only

ID	Name	Specialization	Department	Availability	Action
D001	Dr. Faisal Rahman	Cardiology	Cardiology	Available	Select
D002	Dr. Nusrat Ahmed	Pulmonology	Pulmonology	Available	Select
D003	Dr. Tanvir Hasan	General Medicine	General Medicine	On Leave	Select
D004	Dr. Samia Karim	Internal Medicine	Internal Medicine	Available	Select
D005	Dr. Rafiq Uddin	Neurology	Neurology	Available	Select

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Medicine Stock Management

View, update and get AI-based low stock alerts and restocking suggestions.

Add / Update Medicine

Medicine ID: M010

Medicine Name: Paracetamol 500mg

Category: Pain Reliever

Unit: Tablet

Current Stock: 120

Minimum Stock: 50

Expiry Date: 2026-12-31

+ Add / Update Medicine | Clear

AI Stock Alert & Suggestion

Low Stock Detected!

Medicine: Paracetamol 500mg
Current Stock: 12 tablets
Minimum Stock: 50 tablets
AI Suggestion: Restock at least 100 tablets soon to avoid shortage.

Stock Status: Sufficient

Medicine: Amoxicillin 500mg
Current Stock: 200 capsules
Minimum Stock: 50 capsules
AI Suggestion: Stock level is adequate.

Medicine List

Search medicine name... | All Categories | All Status

ID	Medicine Name	Category	Current Stock	Minimum Stock	Expiry Date	Status	Action
M001	Paracetamol 500mg	Pain Reliever	12	50	2026-12-31	Low Stock	Edit
M002	Amoxicillin 500mg	Antibiotic	200	50	2027-06-15	Sufficient	Edit
M003	Metformin 500mg	Diabetes	30	100	2026-11-20	Low Stock	Edit
M004	Saline (500ml)	IV Fluid	150	50	2027-08-10	Sufficient	Edit
M005	Omeprazole 20mg	Gastric	8	30	2026-09-05	Low Stock	Edit
M006	Cetrizine 10mg	Antihistamine	75	50	2027-03-12	Sufficient	Edit

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Resource Monitoring

Monitor real-time status of beds, medicines, doctors and other hospital resources.

Tuesday, 24 September 2025
12:10 PM

Total Beds
120
Occupied: 96
Available: 24
80% occupied

Total Doctors
22
On Duty: 18
On Leave: 4
82% available

Total Medicines
50
Low Stock: 7
Sufficient: 43
14% low stock

Total Wards
4
Active: 4
Inactive: 0
100% active

Bed Status

View All

Ward	Total Beds	Occupied	Available	Status
General	60	40	20	Moderate
ICU	20	18	2	High
HDU	20	8	12	Normal
Private	20	6	14	Normal

Doctor Status

View All

ID	Name	Specialization	Status
D001	Dr. Faisal Rahman	Cardiology	Available
D002	Dr. Nusrat Ahmed	Pulmonology	Available
D003	Dr. Tanvir Hasan	General Medicine	On Leave
D004	Dr. Samia Karim	Internal Medicine	Available
D005	Dr. Rafiq Uddin	Neurology	Available

Medicine Stock Status

View All

ID	Medicine Name	Current Stock	Minimum Stock	Status
M001	Paracetamol 500mg	12	50	Low Stock
M002	Amoxicillin 500mg	200	50	Sufficient
M003	Metformin 500mg	30	100	Low Stock
M004	Saline (500ml)	150	50	Sufficient
M005	Omeprazole 20mg	8	30	Low Stock

Other Resources

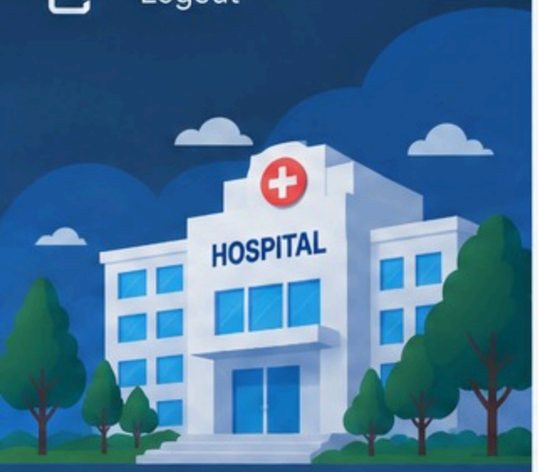
View All

Resource	Total	In Use	Available	Status
Ventilators	10	6	4	Moderate
Oxygen Cylinders	50	30	20	Normal
Wheelchairs	20	12	8	Normal
Monitors	30	22	8	Moderate
Infusion Pumps	15	10	5	Normal

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Demand Prediction

Predict future demand for beds, medicines and doctors using AI analysis.

Date Range
Sep 1, 2025 - Sep 30, 2025

Prediction Type: Bed Demand | Time Period: Next 30 Days | Ward/Category: All | Generate Prediction (AI)

Predicted Bed Demand (Next 30 Days)



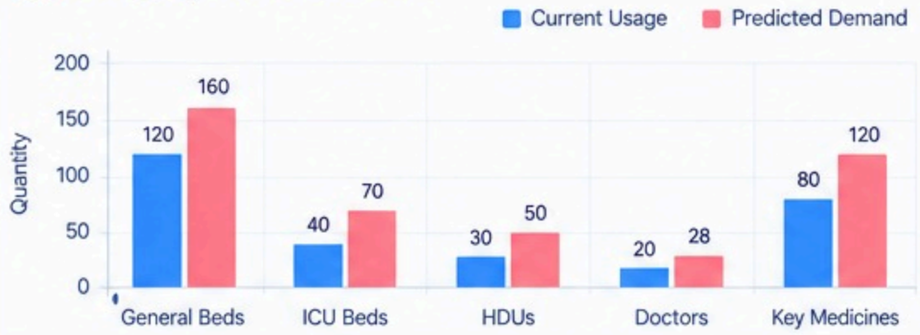
AI Prediction Summary

Expected Increase
+35%

Bed demand is likely to increase in the next 30 days.

Reason:	Seasonal flu cases and higher respiratory infections are expected.
Peak Period:	Sep 20 - Sep 28, 2025
Recommended Action:	Keep extra beds ready in ICU and General Ward.

Category-wise Prediction



Top 5 High-Demand Medicines (Next 30 Days)

Rank	Medicine Name	Current Usage	Predicted Demand	Expected Increase
1	Paracetamol 500mg	500	800	+60%
2	Amoxicillin 500mg	300	450	+50%
3	Saline (500ml)	200	320	+60%
4	Omeprazole 20mg	150	240	+60%
5	Metformin 500mg	100	180	+80%

Recent Prediction History

ID	Type	Time Period	Ward/Category	Predicted Increased	Generated On	Action
P001	Bed Demand	Next 30 Days	All	+35%	Sep 24, 2025 12:10 PM	View
P002	Medicine Demand	Next 30 Days	Antibiotics	+50%	Sep 22, 2025 10:30 AM	View
P003	Doctor Demand	Next 30 Days	Cardiology	+25%	Sep 20, 2025 02:15 PM	View
P004	Bed Demand	Next 7 Days	ICU	+40%	Sep 18, 2025 11:45 AM	View

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AI Insights & Reports

Get AI-powered insights, recommendations and reports for better decision making.

Total Patients
248 ↑ 12% vs last month

Bed Occupancy
80% ↑ 8% vs last month

Low Stock Items
7 ↑ 3 vs last month

Doctors on Duty
18 ↑ 2 vs last month

AI Insights

High Bed Demand Expected
Bed demand is likely to increase by 35% in the next 30 days, especially in ICU and General Ward.

Medicine Restock Recommended
Paracetamol 500mg and Saline (500ml) are expected to run low within 1 week.

Increased Patient Inflow
Patient inflow has increased by 20% compared to last month, mainly due to seasonal flu cases.

Optimize Doctor Scheduling
Cardiology department is overloaded. Consider assigning 1 more doctor during peak hours.

AI Recommendations

1. Keep 10 extra ICU beds ready for next 2 weeks.
2. Restock Paracetamol 500mg (at least 1000 tablets).
3. Assign an additional Cardiologist during peak hours.
4. Monitor medicine usage for seasonal diseases.
5. Improve patient flow management to reduce waiting time.

Generate AI Report

Report Type: Monthly Summary

Month: September 2025

Generate Report (AI)

Recent AI Insights

ID	Insight Type	Summary	Generated On	Action
A001	Bed Demand	Bed demand may increase by 35% in next 30 days.	Sep 24, 2025 12:10 PM	View
A002	Medicine Stock	Paracetamol 500mg stock will be low within 1 week.	Sep 22, 2025 10:30 AM	View
A003	Patient Inflow	Patient inflow increased by 20% compared to last month.	Sep 20, 2025 02:15 PM	View
A004	Doctor Scheduling	Cardiology department is overloaded.	Sep 18, 2025 11:45 AM	View
A005	Resource Usage	ICU bed occupancy reached 90%.	Sep 15, 2025 09:20 AM	View